

13 | OVERVIEW


Coping with Multiple Climate Challenges in the Mystic River Watershed

Abstract

Climate change is a risk multiplier: heat, flooding, chemical pollution, and traffic exhaust can strike the same community at once. Researchers from Harvard Chan C-CHANGE and Boston University are working with residents, local officials, and community groups in the Mystic River Watershed near Boston to identify who is most at risk from these cumulative hazards and which solutions would protect health most.

Introduction

Imagine a flood sweeping through an industrial area, causing stored chemicals to leak into nearby neighborhoods. Residents would face two crises at the same time: the flood itself and exposure to toxic substances. This is what scientists mean by cumulative hazards. The Mystic River Watershed in Massachusetts includes 21 communities, among them parts of Boston, Cambridge, Chelsea, Everett, Revere, and Somerville. A majority of these communities include residents who are low-income and people of color, who often face higher exposure to pollution and climate risks. Researchers want to understand how heat, flooding, chemical exposures from industrial sites, and traffic pollution combine to affect health, and how to build communities that can withstand them.

How Do Scientists Study This?



FROM THE ARTICLE

Project location: Mystic River Watershed, Massachusetts. Map: Harvard Chan C-CHANGE.

The research team works directly with municipal staff, community organizations, and residents across the watershed. Partners include the Mystic River Watershed Association (MyRWA) and the Resilient Mystic Collaborative, which help engage residents and bring forward their priorities and solutions. Researchers map where climate hazards such as heat and flooding overlap with industrial chemical storage and heavy traffic. They then identify which neighborhoods and groups are most at risk from these combined exposures. Finally, they quantify the health benefits of climate resilience solutions so decision makers can prioritize the policies that help the most vulnerable communities first. Results are shared through a public data dashboard.



FROM THE ARTICLE

Community members discuss hazard maps of the watershed. Photo: Muskaan Khemani / BUSPH.

Explore the Evidence

How cumulative hazards build up

1

Extreme heat

Hot days stress bodies and homes

2

Flooding

Water reaches homes and industry

3

Chemical leaks

Floods release stored pollutants

4

Combined harm

Residents face several crises at once

Map the hazards

Researchers identify where heat, flooding, industrial chemicals, and traffic pollution overlap.

Find who is at risk

Data and community input reveal which residents face the greatest combined exposures.

Measure solutions

The team estimates the health benefits of resilience solutions to guide local policy.

What Did They Discover?

This ongoing project is building data-driven tools that local leaders can use to protect people vulnerable to cumulative climate hazards. Its publicly available ACRES data dashboard lets residents, researchers, and policymakers explore cumulative exposures across the watershed. Because the project centers community voices, residents help shape which problems are studied and which solutions are tested. A former EPA regional administrator for New England said the project will yield tangible benefits for people in the watershed and provide policy guidance for the EPA nationwide. By looking at hazards together rather than one at a time, the research can reveal risks that single-hazard studies would miss.

The Mystic River project

21

communities in the watershed study area

4

stressors studied: heat, floods, chemicals, traffic

1

public ACRES dashboard for residents and leaders

A climate resilience manager at MyRWA described the project as blending academic research with lived experience to help people stay safe from chemical exposures.

Takeaways

Risks multiply

Climate hazards can combine with pollution, creating crises bigger than either alone.

Equity matters

Low-income communities and people of color often face the highest combined exposures.

Data for action

Public dashboards help leaders put the most vulnerable communities first.

Key Terms

Watershed

An area of land where all the water drains into the same river, lake, or other body of water.

Cumulative hazards

Multiple threats, such as heat, flooding, and pollution, that affect the same people at the same time.

Risk multiplier

Something that makes existing dangers worse. Climate change multiplies risks from pollution and poverty.

Environmental justice

The fair treatment of all people in environmental decisions, so no group bears an unfair share of pollution or risk.

Data dashboard

An online tool that displays data in maps and charts so people can explore and use it.

Vocabulary Path · Coping with Multiple Climate Challenges in the Mystic River Watershed



Activity

Layered Risk Map

Students layer different hazards on a map to find places where risks pile up.

Materials: Base map of a neighborhood or town; Transparent sheets or tracing paper; Markers in four colors

1. On separate transparent sheets, mark areas at risk of flooding, extreme heat, industrial sites, and busy roads.
2. Stack the sheets on the base map and circle places where three or more hazards overlap.
3. Imagine a flood reaches an industrial site in one circled area. Describe what residents would face.
4. Choose one solution, such as trees for shade or moving chemical storage, and explain which hazards it reduces.
5. Discuss why decision makers should consider hazards together rather than one at a time.

References

Harvard T.H. Chan School of Public Health, C-CHANGE. Helping Communities in the Mystic River Watershed Area Cope with Multiple Climate Challenges.

<https://hsph.harvard.edu/research/climate-health-c-change/research/helping-communities-in-the-mystic-river-watershed-area-cope-with-multiple-climate-challenges/>